

ASSIGNMENT DESCRIPTION

Coffee Export Value Chain Management System (CEVCMS)

Prepared for

Ethiopian Coffee and Tea Authority (ECTA), Addis Ababa, Ethiopia

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Document Type	Assignment Description (Practical Course Deliverable)
Related Documents	Phase 1 Discovery Report (ECTA); SRS ECTA-CEVCMS-SRS-V2.1
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0. Revision History

Version	Date	Author(s)	Description of Changes
0.1	Early Week 1	Project Group	Initial draft prepared from information available at the start of the practical course, the ECTA Phase 1 Discovery Report, and the SRS.
0.2	Mid Week 1	Project Group	Second draft incorporating clarifications obtained from the client interview and comments from the supervisors.
1.0	End of Week 1	Project Group	Final version incorporating written comments from the client and the steering committee.

1. Project Data

Field	Description
Client	Ethiopian Coffee and Tea Authority (ECTA), Addis Ababa, Ethiopia
Name of Project	Coffee Export Value Chain Management System (CEVCMS)
Project Participants	Project group members, contact e-mail addresses and telephone numbers as registered with the practical course administration
Planned Work Load	In accordance with the practical course workload norm for a one-month (four-week) practical course; distributed across analysis, design, implementation, testing, and reporting activities across that period
Document Reference	ECTA-CEVCMS-AD-V1.0, consistent with the SRS reference ECTA-CEVCMS-SRS-V2.1

2. Introduction

This Assignment Description formalizes, from a project management perspective, the scope and terms of the Coffee Export Value Chain Management System (CEVCMS) assignment commissioned by the Ethiopian Coffee and Tea Authority (ECTA). It builds directly on two documents already accepted by the client and the steering committee: the Phase 1 Discovery Report, which establishes the stakeholder landscape and digital-readiness constraints of the Ethiopian coffee export sector, and the Software Requirements Specification (SRS, reference ECTA-CEVCMS-SRS-V2.1), which defines the functional and non-functional requirements of the system to be built.

Ethiopian coffee exports currently move through a sequence of independent actors, from smallholder farmers and primary cooperatives through dry mills, quality inspection, export licensing, banking, and customs, before reaching international buyers. The Discovery Report found that this sequence is presently coordinated through paper ledgers, manual weighbridge slips, and disconnected departmental systems, which limits traceability, slows clearance, and exposes the value chain to price uncertainty and fraud. CEVCMS is ECTA's response: a national digital platform intended to unify these actors under a single, traceable, and auditable workflow, ranging from farm-gate origin to overseas shipment.

This document does not repeat the stakeholder profiles of the Discovery Report or the detailed functional requirements of the SRS. Instead, it defines, for the purposes of the practical course, why the project is

undertaken, what the project group is expected to deliver, how the work will be organized and reported, and how the client will determine that the assignment has been completed satisfactorily.

The SRS describes ECTA's complete enterprise vision for CEVCMS, spanning the full coffee export lifecycle. This Assignment Description defines a carefully selected, one-month-feasible subset of that vision for design and implementation by the project group over the four-week duration of the practical course. The project group is not expected to deliver, and this document does not represent, a national-scale ECTA system; the modules and use cases outside the scope defined in Section 8 remain part of ECTA's longer-term enterprise programme and are explicitly outside the implementation scope of this assignment.

3. Background

Smallholder farmers account for the large majority of Ethiopian coffee production, spread across an estimated four to five million farm holdings, and supply the value chain through primary collectors, cooperatives, and farmers' unions. From there, coffee passes through private and cooperative dry mills, ECTA quality inspection and Q-Grading, export licensing, National Bank of Ethiopia (NBE) foreign exchange clearance, commercial bank letter-of-credit processing, Ethiopian Customs Commission (ECC) clearance, and inland logistics to the Port of Djibouti, before reaching international buyers and roasters.

The Discovery Report characterizes this chain as operationally fragmented: information is recorded on paper at almost every stage, re-entered manually between organizations, and rarely shared in real time between the parties who depend on it. This background motivates the objectives described in Section 5 and constrains the scope described in Section 8.

4. Problem Definition

The Discovery Report's pain-point analysis identifies a consistent set of problems across the value chain, several of which are classified as having critical or high operational impact. These problems, rather than the project group's proposed solution, define the scope of what ECTA is asking to have addressed.

4.1 Component Problems

Problem Area	Description	Priority
Fragmented, paper-based records	Cherry intake, cooperative dividend ledgers, quality certificates, waybills, and customs manifests are recorded and reconciled manually across independent organizations, producing arithmetic errors, lost receipts, and slow reconciliation.	Critical
Lack of end-to-end traceability	No single record links a farmer's plot, a collection batch, a processed lot, and an export shipment, which prevents Ethiopia from demonstrating compliant, traceable origin to buyers and regulators.	Critical
EUDR geolocation compliance gap	Farm boundaries are not captured or verified digitally, creating a risk that export containers are rejected by European Union buyers under the EU Deforestation Regulation (EUDR).	Critical
Foreign exchange and pricing leakage	Export volumes, minimum contract pricing, and commercial bank forex settlement are cross-checked manually, allowing inconsistencies between declared and settled values.	High

Problem Area	Description	Priority
Transit and logistics opacity	Truck movements between dry mills, Addis Ababa warehouses, and the Port of Djibouti are not tracked in real time, leaving exporters and customs unable to verify transit integrity.	High
Price uncertainty for smallholders	Farmers lack direct, timely visibility of market prices, which reduces trust in the value chain and encourages sales to informal traders outside the regulated system.	Medium
Long clearance lead times	Quality grading, waybill issuing, and export clearance currently take substantially longer than the target the client has set, delaying shipment and increasing cost.	High

These problems are discussed in full, with severity ratings and root causes, in Part 5 of the Discovery Report (“Comprehensive Pain Point Analysis Matrix”), to which this description defers for supporting detail. The project group is not expected to resolve every listed problem; Section 5 defines which of these problems the project targets and how success will be measured.

5. Project Objectives

The objectives below are derived from the business objectives and project goals already agreed with ECTA in the SRS (Sections 1.3 and 1.4) and are restated here in SMART form for the purposes of this assignment.

#	Objective	Measure of Success
O1	Replace paper-based waybills, grading certificates, and export permits with cryptographically verifiable digital records.	100% of newly issued waybills, grade certificates, and export permits are generated and verified digitally within the system.
O2	Provide origin traceability for exported coffee lots down to the farmer plot or primary collection batch.	At least five representative export lots processed within the pilot scope can be traced end to end to their originating farm polygon or collection batch on request (exact pilot count to be confirmed with the client at Milestone M3).
O3	Enforce automated cross-validation between export volume, minimum contract pricing, and bank forex settlement.	All discrepancies present in the pilot test dataset between declared export value and settled forex are flagged automatically by the system, with no manual reconciliation step required for the pilot scenarios tested. Delivery of O3 is treated as stretch scope within the one-month timeframe (see Section 8.1).
O4	Reduce the time required for quality grading, waybill issuing, and export clearance.	Measured clearance time for at least three pilot transactions is reduced from the current paper-based baseline toward the 48-hour target set by ECTA. Delivery of O4 is treated as stretch scope within the one-month timeframe (see Section 8.1).
O5	Give smallholder farmers direct visibility of buying prices and digital proof of cherry delivery.	Daily price information and digital receipts are available to farmers through SMS/USDD for at least one pilot cooperative (exact number of participating farmers to be confirmed with the client at Milestone M3).

Consistent with the guidance in Appendix 1, it is not possible to define at the start of the project precisely which of these targets will be fully realized within the one-month practical-course timeframe. Objectives O1 and O2 are treated as core scope, to be delivered within the pilot modules identified in Section 8.1; Objectives O3 and O4 are treated as stretch scope, to be attempted only after the core scope is implemented and tested; Objective O5 is delivered to the extent that the Authentication and Farmer and Polygon Registry modules support it. The assignment description therefore allows for a refinement of scope, in agreement with the client and supervisors, once the analysis and design activities completed by Milestone M3 (Section 9) have produced a more detailed picture of what can be delivered.

6. Project Vision to Problem Definition and Objectives

The project group's approach follows the multi-tier technology strategy already validated with ECTA in the Discovery Report: USSD and SMS interaction for smallholder farmers with basic feature phones, offline-first Android applications for cooperative agents, dry-mill operators, and logistics actors operating in areas with intermittent connectivity, and web-based portals and secure APIs for exporters, banks, and government regulators. This strategy, and the module architecture that implements it, is specified in full in the SRS and is treated here as a fixed constraint rather than a design decision to be revisited.

Given the scope of a practical-course assignment, the project group will adopt an incremental, requirements-driven development approach: work will proceed module by module, following the priority ordering already established in the SRS module directory and requirement traceability matrix, so that the highest-priority regulatory and traceability functions are addressed first and lower-priority analytics and future-enhancement functions are addressed only if time allows. This approach is chosen because the Discovery Report and SRS jointly represent a large, multi-year enterprise programme, of which the practical-course project will realize a bounded, well-defined subset: the Authentication and Role-Based Access, Farmer and Polygon Registry, and Traceability and QR Engine modules, as identified in Section 8.1.

7. Planning

7.1 Deliverables

Deliverable	Description
Assignment Description	This document; defines the problem, objectives, scope, and terms of the assignment, to be agreed with the client and the steering committee.
Project Plan	Internal planning document (per Appendix 2) detailing activities, resource allocation, and quality procedures for the project group.
Design Document	Elaboration of the subset of SRS modules selected for implementation, including data model, interfaces, and module interaction diagrams consistent with the SRS module directory.
Working Software Increment	An implemented and tested subset of CEVCMS functionality, scoped from the SRS functional requirements, demonstrating the selected use cases end to end.
Test Report	Evidence of unit, integration, and (where applicable) offline-synchronization testing of the delivered increment.
Progress Reports	Biweekly reports to the supervisor per Appendix 3.
Final Presentation and Demonstration	A presentation to the client and steering committee demonstrating the delivered increment against the objectives in Section 5.

7.2 Project Activities

The activities below give a concrete, high-level vision of how the objectives in Section 5 will be pursued. They are ordered to align with the project's phased approach and the module priority already established in the SRS.

Activity	Relates to
Confirm assignment scope with client and steering committee, including which SRS modules and use cases fall within the practical-course boundary.	O1–O5, Scope (Sec. 8)
Review Discovery Report stakeholder profiles and SRS module directory to select the pilot user classes and modules for implementation.	Vision (Sec. 6)
Elaborate the design for the selected modules (data model, interfaces, offline/online behavior) consistent with the SRS architecture.	O1, O2
Implement the selected functional requirements, prioritizing the core regulatory and traceability modules (Authentication and Role-Based Access; Farmer and Polygon Registry; Traceability and QR Engine) ahead of the stretch-scope modules (Quality Grading and Waybill Issuance; Export–Forex Cross-Validation).	O1, O2, O4
Implement automated cross-validation logic between export, pricing, and settlement data where in scope.	O3
Test the implemented increment, including applicable offline-synchronization scenarios.	O1–O5
Prepare and deliver progress reports and the final demonstration to the client.	Progress Reporting (Sec. 10)

8. Project Scope

CEVCMS as specified in the SRS spans eighteen functional modules across the full upstream-to-downstream coffee export lifecycle, together with national-scale infrastructure, multi-region deployment, and integrations with eight external institutions including the National Bank of Ethiopia, the Ethiopian Customs Commission, commercial banks, and telecom providers. This full scope constitutes ECTA's enterprise programme and is not, by itself, achievable within a practical-course timeframe.

The scope of this assignment is therefore a bounded subset of the SRS, to be selected jointly by the project group, the supervisors, and the client during the second week of the project, in line with the procedure described in Appendix 1. Selection will be guided by the module priority ordering and phased implementation roadmap already agreed with ECTA (Discovery Report, Part 10), which places core regulatory infrastructure, farmer and polygon registry, quality grading, and traceability ahead of analytics, future EUDR roadmap items, and full multi-region production infrastructure.

8.1 In Scope (subject to confirmation with the client)

- Analysis, design, and implementation of the following SRS modules, planned as the core implementation scope, subject to confirmation with the client and supervisors at Milestone M3 (Section 9): (1) Authentication and Role-Based Access; (2) Farmer and Polygon Registry; (3) Traceability and QR Engine. If time allows after the core scope is implemented and tested, the Quality Grading and Waybill Issuance module and the Export–Forex Cross-Validation module are targeted as stretch scope, supporting Objectives O4 and O3 respectively.
- Implementation of the corresponding use cases at a scale suitable for demonstration, rather than national production volume.

- A representative subset of the external interfaces defined in the SRS, implemented as simulated or sandboxed integrations rather than live connections to production institutional systems.
- Testing of the delivered increment against the acceptance criteria defined in Section 12.

8.2 Out of Scope

- Full production deployment on the multi-region Kubernetes infrastructure specified in Section 14 of the SRS.
- Live integration with production systems of the National Bank of Ethiopia, the Ethiopian Customs Commission, commercial banks, or telecom carriers; these will be represented by stubs or sandbox equivalents unless the client makes production access available.
- Future-enhancement items listed in the SRS roadmap (AI-based defect analysis, satellite yield estimation, blockchain smart contracts, IoT warehouse sensors).
- National-scale performance and load targets (e.g., peak-harvest throughput); the delivered increment will instead be tested at a scale appropriate to a course project.

9. Milestones

Milestone	Planned Timing	Description	Associated Deliverable(s)
M1 – Assignment Description Agreed	End of Week 1	First draft reviewed with client and supervisors; open questions raised in the client interview and resolved; final Assignment Description signed off.	Assignment Description
M2 – Project Plan Approved	End of Week 1	Project plan finalized in consultation with supervisors, consistent with this Assignment Description.	Project Plan
M3 – Scope and Design Finalized	Mid Week 2	Pilot modules and use cases confirmed with the client; design document completed for the selected scope.	Design Document
M4 – Working Increment Delivered	End of Week 3	Selected functional requirements implemented and internally tested.	Working Software Increment, Test Report
M5 – Client Acceptance Review	Mid Week 4	Delivered increment demonstrated to the client against the acceptance procedure in Section 12.	Test Report, Final Presentation
M6 – Project Closure	End of Week 4	Final documentation delivered; project formally closed with client and supervisors.	Final Presentation and Demonstration

Exact calendar dates will be fixed in the Project Plan (Appendix 2) once the practical course schedule is confirmed; the week numbers above are relative to the start of the course.

10. Overall Planning

The overall planning integrates the deliverables, activities, and milestones defined above into a single timeline, to be elaborated with dates and resource assignments in the Project Plan, in accordance with

Appendix 2. The Project Plan will use the deliverables of Section 7.1 and the activity list of Section 7.2 as its basis, and will remain subject to revision during the project, whereas this Assignment Description, once agreed with the client, will not be revised except in exceptional circumstances.

11. Progress Reporting

The project group will send a biweekly progress report to the supervisor by e-mail, following the format defined in Appendix 3: project name, date, author, report period, work completed, deviations from planning and their consequences, bottlenecks and how they are addressed, planned activities for the next period, and a self-assessment. In addition, an informal status update will be shared with the ECTA client contact at each major milestone listed in Section 9, and a formal demonstration will be given at project closure.

12. Acceptance Procedure

At the end of the project, the client will determine whether the delivered increment meets the objectives defined in Section 5, within the scope defined in Section 8. Acceptance will be assessed as follows:

Step	Description
1. Demonstration	The project group demonstrates the delivered use cases end to end, using the acceptance criteria already defined per functional requirement in the SRS (e.g., authentication response time, farmer registration validation, polygon EUDR flagging) as the basis for evaluation.
2. Traceability Check	The client verifies that at least one representative coffee lot can be traced from farm-level origin (or collection batch) through to the point reached within the implemented scope, consistent with Objective O2.
3. Test Report Review	The client and supervisors review the Test Report for evidence that the implemented functional requirements have been verified.
4. Sign-Off	The client confirms, in writing, whether the delivered increment is accepted, accepted with minor remarks, or requires rework, referencing the specific objectives or requirements not met.

13. Agreements on Cooperation

- Language: project communication and documentation will be conducted in English; user-facing interface text, where implemented, will follow the SRS requirement for Amharic, Afaan Oromo, Tigrinya, and English support only to the extent in scope.
- Client contact: the project group will communicate with ECTA through a designated client contact point, to be confirmed at the Week 1 client interview.
- Access to information: the Discovery Report and SRS constitute the authoritative source of stakeholder and requirement information; any additional information needed will be requested from the client contact rather than assumed.
- Tools and infrastructure: development tools, source control, and any sandboxed integration environments will be agreed with the supervisors at project start.

14. Additional Conditions / Assumptions

The following assumptions and constraints, drawn directly from the SRS, apply to this assignment and are not altered by it:

- Regulatory compliance: any implemented functionality must remain consistent with Ethiopian Coffee Export Directives, NBE Forex Regulations, and EUDR (Regulation EU 2023/1115), even where full compliance validation is out of the practical-course scope.
- Data sovereignty: any farmer, financial, or trade data used or generated during the project will be treated as if subject to the SRS requirement that such data reside within Ethiopia, and no real personal or financial data will be used outside a sandboxed environment without explicit client authorization.
- Connectivity assumption: smallholder farmers are assumed to have access to at least a basic GSM feature phone capable of SMS or USSD, consistent with the SRS assumption.
- Dependency: any demonstration of USSD/SMS functionality depends on either a live telecom gateway being made available by ECTA or the use of a simulated gateway agreed with the supervisors.
- Risk carried forward from the SRS Risk Analysis: rural network blackout, collection-agent fraud, EUDR mapping error, and e-Waybill falsification are recognized as high-impact risks for the full CEVCMS programme; where the practical-course scope touches these areas, the corresponding SRS mitigation (offline queuing, Bluetooth scale integration, PostGIS polygon validation, HMAC-signed QR codes) will be treated as a design requirement rather than re-derived independently.

15. References

- Ethiopian Coffee and Tea Authority (ECTA). Phase 1: User Research, Stakeholder Profiling, & Digital Readiness Report — Coffee Export Value Chain Management System.
- Software Requirements Specification (SRS) for the Coffee Export Value Chain Management System (CEVCMS), Document Reference ECTA-CEVCMS-SRS-V2.1, Version 2.1.0.
- University Practical Course, Appendix 1 – Assignment Description.
- University Practical Course, Appendix 2 – Minimum Project Plan.
- University Practical Course, Appendix 3 – Minimum Progress Report.
- ISO/IEC/IEEE 29148:2018, Systems and software engineering — Life cycle processes — Requirements engineering.
- European Union Regulation (EU) 2023/1115 on Deforestation-free Products (EUDR).